

| Mapped CO(s) Number | Practical/ Lab Session Outcomes (LSOs)                                                                                                                                                                                       | Laboratory Experiment/ Practical Titles                                                                 |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| CO1                 | <i>LSO 1.1</i> Identify active and passive components used in electrical circuits.<br><i>LSO 1.2</i> Verify Ohm's Law for a resistive load.                                                                                  | LE- 1 Identification of active and passive components and Ohm's Law verification                        |
| CO1                 | <i>LSO 2.1</i> Analyse series and parallel circuits to verify Kirchhoff's Voltage and Current Laws.<br><i>LSO 2.2</i> Apply voltage division and current division rules to determine circuit parameters experimentally.      | LE- 2 Application of Kirchhoff's Laws in series and parallel circuits                                   |
| CO2                 | <i>LSO 3.1</i> Identify the effect of low power factor in a single-phase load.<br><i>LSO 3.2</i> Demonstrate power factor improvement of a single-phase load using a capacitor.                                              | LE- 3 Improvement of power factor of a single-phase load using a capacitor                              |
| CO3                 | <i>LSO 4.1</i> Measure primary and secondary voltages of a single-phase transformer under test conditions.<br><i>LSO 4.2</i> Verify the turns ratio by comparing primary and secondary winding values.                       | LE- 4 Determination of voltage and turns ratios of a single-phase transformer                           |
| CO3                 | <i>LSO 5.1</i> Identify the main constructional parts of DC machines and induction machines using cut-section models.<br><i>LSO 5.2</i> Examine the functions of different components of DC machines and induction machines. | LE- 5 Identification of main parts of DC machines and induction machines using cut-section models       |
| CO4                 | <i>LSO 6.1</i> Identify the behavior of a P-N junction diode under forward and reverse bias conditions.<br><i>LSO 6.2</i> Analyse the V-I characteristics of the diode from observed readings.                               | LE- 6 Determination of V-I characteristics of a P-N junction diode                                      |
| CO4                 | <i>LSO 7.1</i> Verify half-wave and full wave rectifier circuits with and without filters.<br><i>LSO 7.2</i> Compare the output performance of rectifiers using sinusoidal input under filtered and unfiltered conditions    | LE- 7 Performance of half-wave and full-wave rectifiers with and without filters using sinusoidal input |
| CO4                 | <i>LSO 8.1</i> Identify the components and circuit setup of a Zener diode voltage regulator.<br><i>LSO 8.2</i> Analyse the regulation performance of the Zener diode under varying input voltage or load conditions.         | LE- 8 Analysis of the operation of a voltage regulator using a Zener Diode                              |

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| CO4                        | <p><i>LSO 9.1</i> Identify the input and output characteristics of a common emitter (CE) bipolar junction transistor.</p> <p><i>LSO 9.2</i> Analyse the variation of current and voltage in CE transistor operation from observed characteristics.</p>                                    | LE- 9 Input and output characterization of common emitter (CE) bipolar junction transistor          |
| CO5                        | <p><i>LSO 10.1</i> Construct basic logic gates (AND, OR, NOT, XOR) using only NAND and NOR gates.</p> <p><i>LSO 10.2</i> Verify the output of implemented gates using truth tables.</p>                                                                                                   | LE- 10 Implementation of logic gates (AND, OR, NOT, XOR) using Universal gates                      |
| CO6                        | <p><i>LSO 11.1</i> Measure amplitude and time period of signals using a Cathode Ray Oscilloscope (CRO).</p> <p><i>LSO 11.2</i> Determine signal frequency from CRO waveform observations.</p>                                                                                             | LE- 11 Measurement of signal parameters using a CRO                                                 |
| CO6                        | <p><i>LSO 12.1</i> Identify different types of electrical cables, wires, and fuses based on their construction, rating, and application.</p> <p><i>LSO 12.2</i> Demonstrate proper selection and safe handling of cables, wires, and fuses for basic electrical circuit requirements.</p> | LE- 12 Identification of different types of cable, wires and fuses                                  |
| CO6                        | <p><i>LSO 13.1</i> Identify the construction and functions of fuse, MCB, and MCCB.</p> <p><i>LSO 13.2</i> Evaluate the operation of fuse, MCB, and MCCB under fault and overload conditions.</p>                                                                                          | LE- 13 Operation of protection devices such as fuse, MCB, and MCCB                                  |
| CO6                        | <p><i>LSO 14.1</i> Demonstrate proper earthing techniques for electrical installations.</p> <p><i>LSO 14.2</i> Apply electrical safety practices while handling equipment and circuits.</p>                                                                                               | LE- 14 Demonstration of earthing techniques and electrical safety practices                         |
| CO6                        | <p><i>LSO 15.1</i> Implement domestic wiring for a single lamp and fan with regulator using switches.</p> <p><i>LSO 15.2</i> Demonstrate fuse/MCB protection for safe household electrical circuits.</p>                                                                                  | LE- 15 Wiring of single lamp and fan with regulator controlled by a switch with fuse/MCB protection |
| CO6                        | <p><i>LSO 16.1</i> Demonstrate the safe handling and operation of a Megger for insulation and earth resistance testing.</p> <p><i>LSO 16.2</i> Measure earth resistance accurately and interpret the results for effective grounding performance.</p>                                     | LE- 16 Handling of Meggers and measurement of earth resistance.                                     |